

Termination: The loop makes exactly $\lfloor n/2 \rfloor$ iterations, and so it terminates. At termination, $i = 0$. By the loop invariant, each node $1, 2, \dots, n$ is the root of a max-heap. In particular, node 1 is.

We can compute a simple upper bound on the running time of BUILD-MAX-HEAP as follows. Each call to MAX-HEAPIFY costs $O(\lg n)$ time, and BUILD-MAX-HEAP makes $O(n)$ such calls. Thus, the running time is $O(n \lg n)$. This upper bound, though correct, is not as tight as it can be.

We can derive a tighter asymptotic bound by observing that the time for MAX-HEAPIFY to run at a node varies with the height of the node in the tree, and that the heights of most nodes are small. Our tighter analysis relies on the properties that an n -element heap has height $\lfloor \lg n \rfloor$ (see Exercise 6.1-2) and at most $\lceil n/2^{h+1} \rceil$ nodes of any height h (see Exercise 6.3-4).

The time required by MAX-HEAPIFY when called on a node of height h is $O(h)$. Letting c be the constant implicit in the asymptotic notation, we can express the total cost of BUILD-MAX-HEAP as being bounded from above by $\sum_{h=0}^{\lfloor \lg n \rfloor} \lceil n/2^{h+1} \rceil ch$. As Exercise 6.3-2 shows, we have $\lceil n/2^{h+1} \rceil \geq 1/2$ for $0 \leq h \leq \lfloor \lg n \rfloor$. Since $\lceil x \rceil \leq 2x$ for any $x \geq 1/2$, we have $\lceil n/2^{h+1} \rceil \leq n/2^h$. We thus obtain

$$\begin{aligned}
 \sum_{h=0}^{\lfloor \lg n \rfloor} \left\lceil \frac{n}{2^{h+1}} \right\rceil ch &\leq \sum_{h=0}^{\lfloor \lg n \rfloor} \frac{n}{2^h} ch \\
 &= cn \sum_{h=0}^{\lfloor \lg n \rfloor} \frac{h}{2^h} \\
 &\leq cn \sum_{h=0}^{\infty} \frac{h}{2^h} \\
 &\leq cn \cdot \frac{1/2}{(1 - 1/2)^2} \quad (\text{by equation (A.11) on page 1142 with } x = 1/2) \\
 &= O(n).
 \end{aligned}$$

Hence, we can build a max-heap from an unordered array in linear time.

To build a min-heap, use the procedure BUILD-MIN-HEAP, which is the same as BUILD-MAX-HEAP but with the call to MAX-HEAPIFY in line 3 replaced by a call to MIN-HEAPIFY (see Exercise 6.2-3). BUILD-MIN-HEAP produces a min-heap from an unordered linear array in linear time.

Exercises**6.3-1**

Using Figure 6.3 as a model, illustrate the operation of BUILD-MAX-HEAP on the array $A = \langle 5, 3, 17, 10, 84, 19, 6, 22, 9 \rangle$.

6.3-2

Show that $\lceil n/2^{h+1} \rceil \geq 1/2$ for $0 \leq h \leq \lfloor \lg n \rfloor$.

6.3-3

Why does the loop index i in line 2 of BUILD-MAX-HEAP decrease from $\lfloor n/2 \rfloor$ to 1 rather than increase from 1 to $\lfloor n/2 \rfloor$?

6.3-4

Show that there are at most $\lceil n/2^{h+1} \rceil$ nodes of height h in any n -element heap.

6.4 The heapsort algorithm

The heapsort algorithm, given by the procedure HEAPSORT, starts by calling the BUILD-MAX-HEAP procedure to build a max-heap on the input array $A[1:n]$. Since the maximum element of the array is stored at the root $A[1]$, HEAPSORT can place it into its correct final position by exchanging it with $A[n]$. If the procedure then discards node n from the heap—and it can do so by simply decrementing $A.\text{heap-size}$ —the children of the root remain max-heaps, but the new root element might violate the max-heap property. To restore the max-heap property, the procedure just calls MAX-HEAPIFY($A, 1$), which leaves a max-heap in $A[1:n-1]$. The HEAPSORT procedure then repeats this process for the max-heap of size $n-1$ down to a heap of size 2. (See Exercise 6.4-2 for a precise loop invariant.)

```

HEAPSORT( $A, n$ )
1  BUILD-MAX-HEAP( $A, n$ )
2  for  $i = n$  downto 2
3      exchange  $A[1]$  with  $A[i]$ 
4       $A.\text{heap-size} = A.\text{heap-size} - 1$ 
5      MAX-HEAPIFY( $A, 1$ )

```

Figure 6.4 shows an example of the operation of HEAPSORT after line 1 has built the initial max-heap. The figure shows the max-heap before the first iteration of the **for** loop of lines 2–5 and after each iteration.

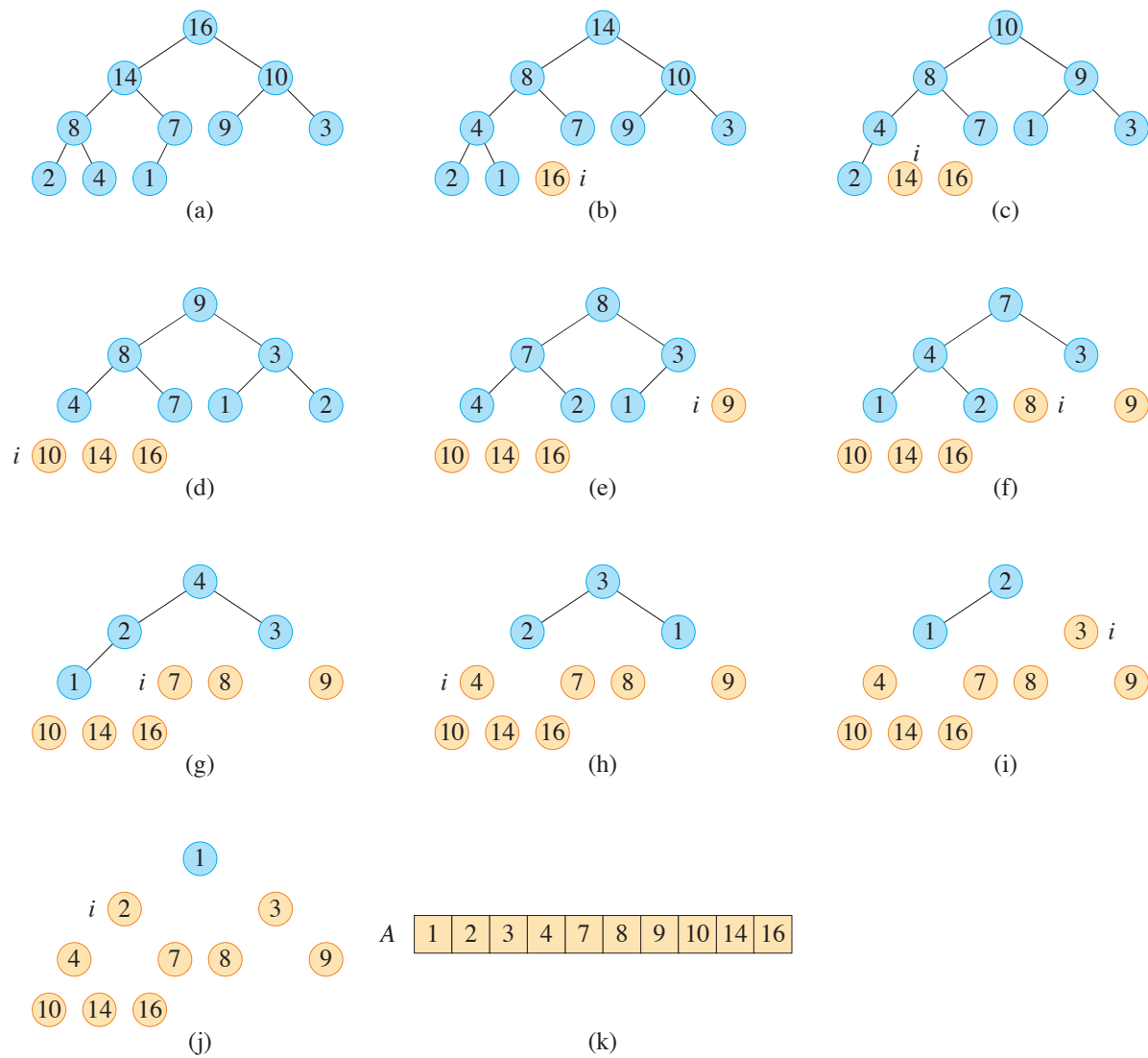


Figure 6.4 The operation of HEAPSORT. (a) The max-heap data structure just after BUILD-MAX-HEAP has built it in line 1. (b)–(j) The max-heap just after each call of MAX-HEAPIFY in line 5, showing the value of i at that time. Only blue nodes remain in the heap. Tan nodes contain the largest values in the array, in sorted order. (k) The resulting sorted array A .

The HEAPSORT procedure takes $O(n \lg n)$ time, since the call to BUILD-MAX-HEAP takes $O(n)$ time and each of the $n - 1$ calls to MAX-HEAPIFY takes $O(\lg n)$ time.

Exercises

6.4-1

Using Figure 6.4 as a model, illustrate the operation of HEAPSORT on the array $A = \langle 5, 13, 2, 25, 7, 17, 20, 8, 4 \rangle$.

6.4-2

Argue the correctness of HEAPSORT using the following loop invariant:

At the start of each iteration of the **for** loop of lines 2–5, the subarray $A[1 : i]$ is a max-heap containing the i smallest elements of $A[1 : n]$, and the subarray $A[i + 1 : n]$ contains the $n - i$ largest elements of $A[1 : n]$, sorted.

6.4-3

What is the running time of HEAPSORT on an array A of length n that is already sorted in increasing order? How about if the array is already sorted in decreasing order?

6.4-4

Show that the worst-case running time of HEAPSORT is $\Omega(n \lg n)$.

★ 6.4-5

Show that when all the elements of A are distinct, the best-case running time of HEAPSORT is $\Omega(n \lg n)$.

6.5 Priority queues

In Chapter 8, we will see that any comparison-based sorting algorithm requires $\Omega(n \lg n)$ comparisons and hence $\Omega(n \lg n)$ time. Therefore, heapsort is asymptotically optimal among comparison-based sorting algorithms. Yet, a good implementation of quicksort, presented in Chapter 7, usually beats it in practice. Nevertheless, the heap data structure itself has many uses. In this section, we present one of the most popular applications of a heap: as an efficient priority queue. As with heaps, priority queues come in two forms: max-priority queues and min-priority queues. We'll focus here on how to implement max-priority queues, which are in turn based on max-heaps. Exercise 6.5-3 asks you to write the procedures for min-priority queues.

A *priority queue* is a data structure for maintaining a set S of elements, each with an associated value called a *key*. A *max-priority queue* supports the following operations:

INSERT(S, x, k) inserts the element x with key k into the set S , which is equivalent to the operation $S = S \cup \{x\}$.

MAXIMUM(S) returns the element of S with the largest key.

EXTRACT-MAX(S) removes and returns the element of S with the largest key.

INCREASE-KEY(S, x, k) increases the value of element x 's key to the new value k , which is assumed to be at least as large as x 's current key value.

Among their other applications, you can use max-priority queues to schedule jobs on a computer shared among multiple users. The max-priority queue keeps track of the jobs to be performed and their relative priorities. When a job is finished or interrupted, the scheduler selects the highest-priority job from among those pending by calling EXTRACT-MAX. The scheduler can add a new job to the queue at any time by calling INSERT.

Alternatively, a *min-priority queue* supports the operations INSERT, MINIMUM, EXTRACT-MIN, and DECREASE-KEY. A min-priority queue can be used in an event-driven simulator. The items in the queue are events to be simulated, each with an associated time of occurrence that serves as its key. The events must be simulated in order of their time of occurrence, because the simulation of an event can cause other events to be simulated in the future. The simulation program calls EXTRACT-MIN at each step to choose the next event to simulate. As new events are produced, the simulator inserts them into the min-priority queue by calling INSERT. We'll see other uses for min-priority queues, highlighting the DECREASE-KEY operation, in Chapters 21 and 22.

When you use a heap to implement a priority queue within a given application, elements of the priority queue correspond to objects in the application. Each object contains a key. If the priority queue is implemented by a heap, you need to determine which application object corresponds to a given heap element, and vice versa. Because the heap elements are stored in an array, you need a way to map application objects to and from array indices.

One way to map between application objects and heap elements uses *handles*, which are additional information stored in the objects and heap elements that give enough information to perform the mapping. Handles are often implemented to be opaque to the surrounding code, thereby maintaining an abstraction barrier between the application and the priority queue. For example, the handle within an application object might contain the corresponding index into the heap array. But since only the code for the priority queue accesses this index, the index is entirely hidden from the application code. Because heap elements change locations within

the array during heap operations, an actual implementation of the priority queue, upon relocating a heap element, must also update the array indices in the corresponding handles. Conversely, each element in the heap might contain a pointer to the corresponding application object, but the heap element knows this pointer as only an opaque handle and the application maps this handle to an application object. Typically, the worst-case overhead for maintaining handles is $O(1)$ per access.

As an alternative to incorporating handles in application objects, you can store within the priority queue a mapping from application objects to array indices in the heap. The advantage of doing so is that the mapping is contained entirely within the priority queue, so that the application objects need no further embellishment. The disadvantage lies in the additional cost of establishing and maintaining the mapping. One option for the mapping is a hash table (see Chapter 11).¹ The added expected time for a hash table to map an object to an array index is just $O(1)$, though the worst-case time can be as bad as $\Theta(n)$.

Let's see how to implement the operations of a max-priority queue using a max-heap. In the previous sections, we treated the array elements as the keys to be sorted, implicitly assuming that any satellite data moved with the corresponding keys. When a heap implements a priority queue, we instead treat each array element as a pointer to an object in the priority queue, so that the object is analogous to the satellite data when sorting. We further assume that each such object has an attribute *key*, which determines where in the heap the object belongs. For a heap implemented by an array A , we refer to $A[i].key$.

The procedure MAX-HEAP-MAXIMUM on the facing page implements the MAXIMUM operation in $\Theta(1)$ time, and MAX-HEAP-EXTRACT-MAX implements the operation EXTRACT-MAX. MAX-HEAP-EXTRACT-MAX is similar to the **for** loop body (lines 3–5) of the HEAPSORT procedure. We implicitly assume that MAX-HEAPIFY compares priority-queue objects based on their *key* attributes. We also assume that when MAX-HEAPIFY exchanges elements in the array, it is exchanging pointers and also that it updates the mapping between objects and array indices. The running time of MAX-HEAP-EXTRACT-MAX is $O(\lg n)$, since it performs only a constant amount of work on top of the $O(\lg n)$ time for MAX-HEAPIFY, plus whatever overhead is incurred within MAX-HEAPIFY for mapping priority-queue objects to array indices.

The procedure MAX-HEAP-INCREASE-KEY on page 176 implements the INCREASE-KEY operation. It first verifies that the new key k will not cause the key in the object x to decrease, and if there is no problem, it gives x the new key value. The procedure then finds the index i in the array corresponding to object x ,

¹ In Python, dictionaries are implemented with hash tables.

```

MAX-HEAP-MAXIMUM(A)
1  if A.heap-size < 1
2      error “heap underflow”
3  return A[1]

MAX-HEAP-EXTRACT-MAX(A)
1  max = MAX-HEAP-MAXIMUM(A)
2  A[1] = A[A.heap-size]
3  A.heap-size = A.heap-size − 1
4  MAX-HEAPIFY(A, 1)
5  return max

```

so that $A[i]$ is x . Because increasing the key of $A[i]$ might violate the max-heap property, the procedure then, in a manner reminiscent of the insertion loop (lines 5–7) of INSERTION-SORT on page 19, traverses a simple path from this node toward the root to find a proper place for the newly increased key. As MAX-HEAP-INCREASE-KEY traverses this path, it repeatedly compares an element’s key to that of its parent, exchanging pointers and continuing if the element’s key is larger, and terminating if the element’s key is smaller, since the max-heap property now holds. (See Exercise 6.5-7 for a precise loop invariant.) Like MAX-HEAPIFY when used in a priority queue, MAX-HEAP-INCREASE-KEY updates the information that maps objects to array indices when array elements are exchanged. Figure 6.5 shows an example of a MAX-HEAP-INCREASE-KEY operation. In addition to the overhead for mapping priority queue objects to array indices, the running time of MAX-HEAP-INCREASE-KEY on an n -element heap is $O(\lg n)$, since the path traced from the node updated in line 3 to the root has length $O(\lg n)$.

The procedure MAX-HEAP-INSERT on the next page implements the INSERT operation. It takes as inputs the array A implementing the max-heap, the new object x to be inserted into the max-heap, and the size n of array A . The procedure first verifies that the array has room for the new element. It then expands the max-heap by adding to the tree a new leaf whose key is $-\infty$. Then it calls MAX-HEAP-INCREASE-KEY to set the key of this new element to its correct value and maintain the max-heap property. The running time of MAX-HEAP-INSERT on an n -element heap is $O(\lg n)$ plus the overhead for mapping priority queue objects to indices.

In summary, a heap can support any priority-queue operation on a set of size n in $O(\lg n)$ time, plus the overhead for mapping priority queue objects to array indices.

MAX-HEAP-INCREASE-KEY(A, x, k)

```

1  if  $k < x.key$ 
2      error “new key is smaller than current key”
3   $x.key = k$ 
4  find the index  $i$  in array  $A$  where object  $x$  occurs
5  while  $i > 1$  and  $A[\text{PARENT}(i)].key < A[i].key$ 
6      exchange  $A[i]$  with  $A[\text{PARENT}(i)]$ , updating the information that maps
        priority queue objects to array indices
7       $i = \text{PARENT}(i)$ 

```

MAX-HEAP-INSERT(A, x, n)

```

1  if  $A.heap\text{-}size == n$ 
2      error “heap overflow”
3   $A.heap\text{-}size = A.heap\text{-}size + 1$ 
4   $k = x.key$ 
5   $x.key = -\infty$ 
6   $A[A.heap\text{-}size] = x$ 
7  map  $x$  to index  $heap\text{-}size$  in the array
8  MAX-HEAP-INCREASE-KEY( $A, x, k$ )

```

Exercises

6.5-1

Suppose that the objects in a max-priority queue are just keys. Illustrate the operation of MAX-HEAP-EXTRACT-MAX on the heap $A = \langle 15, 13, 9, 5, 12, 8, 7, 4, 0, 6, 2, 1 \rangle$.

6.5-2

Suppose that the objects in a max-priority queue are just keys. Illustrate the operation of MAX-HEAP-INSERT($A, 10$) on the heap $A = \langle 15, 13, 9, 5, 12, 8, 7, 4, 0, 6, 2, 1 \rangle$.

6.5-3

Write pseudocode to implement a min-priority queue with a min-heap by writing the procedures MIN-HEAP-MINIMUM, MIN-HEAP-EXTRACT-MIN, MIN-HEAP-DECREASE-KEY, and MIN-HEAP-INSERT.

6.5-4

Write pseudocode for the procedure MAX-HEAP-DECREASE-KEY(A, x, k) in a max-heap. What is the running time of your procedure?

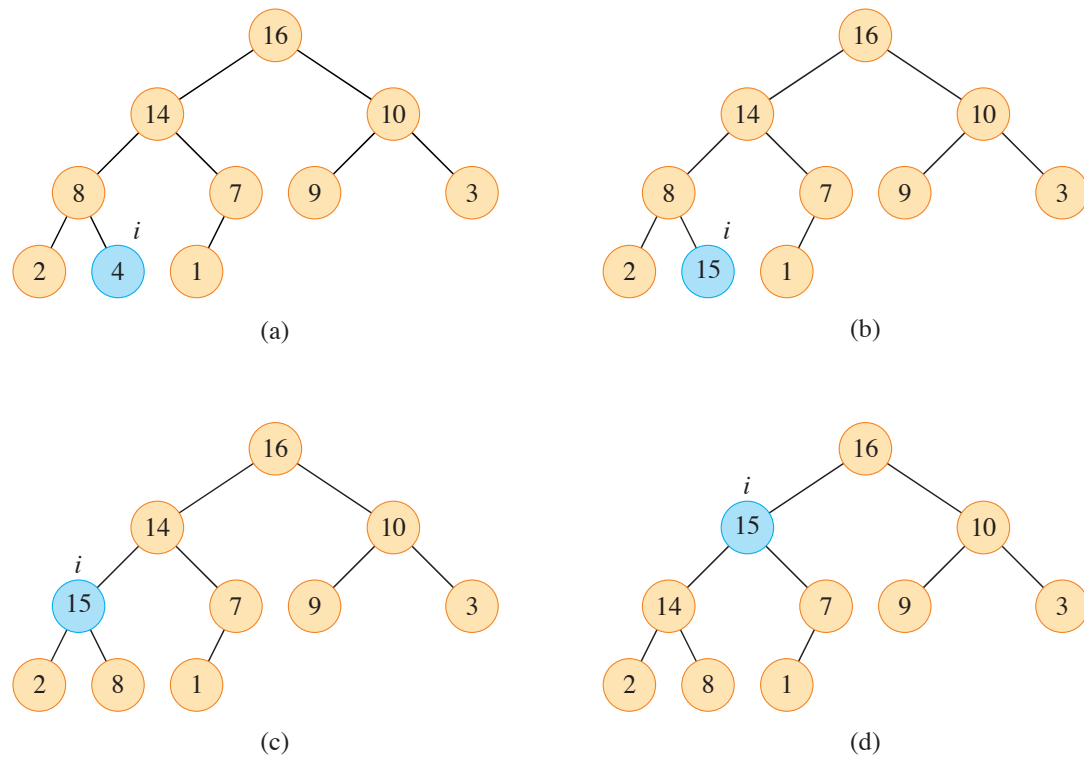


Figure 6.5 The operation of MAX-HEAP-INCREASE-KEY. Only the key of each element in the priority queue is shown. The node indexed by i in each iteration is shown in blue. **(a)** The max-heap of Figure 6.4(a) with i indexing the node whose key is about to be increased. **(b)** This node has its key increased to 15. **(c)** After one iteration of the **while** loop of lines 5–7, the node and its parent have exchanged keys, and the index i moves up to the parent. **(d)** The max-heap after one more iteration of the **while** loop. At this point, $A[\text{PARENT}(i)] \geq A[i]$. The max-heap property now holds and the procedure terminates.

6.5-5

Why does MAX-HEAP-INSERT bother setting the key of the inserted object to $-\infty$ in line 5 given that line 8 will set the object's key to the desired value?

6.5-6

Professor Uriah suggests replacing the **while** loop of lines 5–7 in MAX-HEAP-INCREASE-KEY by a call to MAX-HEAPIFY. Explain the flaw in the professor's idea.

6.5-7

Argue the correctness of MAX-HEAP-INCREASE-KEY using the following loop invariant:

At the start of each iteration of the **while** loop of lines 5–7:

- a. If both nodes $\text{PARENT}(i)$ and $\text{LEFT}(i)$ exist, then $A[\text{PARENT}(i)].\text{key} \geq A[\text{LEFT}(i)].\text{key}$.
- b. If both nodes $\text{PARENT}(i)$ and $\text{RIGHT}(i)$ exist, then $A[\text{PARENT}(i)].\text{key} \geq A[\text{RIGHT}(i)].\text{key}$.
- c. The subarray $A[1 : A.\text{heap-size}]$ satisfies the max-heap property, except that there may be one violation, which is that $A[i].\text{key}$ may be greater than $A[\text{PARENT}(i)].\text{key}$.

You may assume that the subarray $A[1 : A.\text{heap-size}]$ satisfies the max-heap property at the time **MAX-HEAP-INCREASE-KEY** is called.

6.5-8

Each exchange operation on line 6 of **MAX-HEAP-INCREASE-KEY** typically requires three assignments, not counting the updating of the mapping from objects to array indices. Show how to use the idea of the inner loop of **INSERTION-SORT** to reduce the three assignments to just one assignment.

6.5-9

Show how to implement a first-in, first-out queue with a priority queue. Show how to implement a stack with a priority queue. (Queues and stacks are defined in Section 10.1.3.)

6.5-10

The operation **MAX-HEAP-DELETE**(A, x) deletes the object x from max-heap A . Give an implementation of **MAX-HEAP-DELETE** for an n -element max-heap that runs in $O(\lg n)$ time plus the overhead for mapping priority queue objects to array indices.

6.5-11

Give an $O(n \lg k)$ -time algorithm to merge k sorted lists into one sorted list, where n is the total number of elements in all the input lists. (*Hint:* Use a min-heap for k -way merging.)

Problems

6-1 Building a heap using insertion

One way to build a heap is by repeatedly calling **MAX-HEAP-INSERT** to insert the elements into the heap. Consider the procedure **BUILD-MAX-HEAP'** on the facing page. It assumes that the objects being inserted are just the heap elements.